

Saad Driouech

Nuremberg, Germany (Open to relocate) | Available full-time from July 2026

 saad-driouech |  saad-driouech |  saaddriouech.com



Profile Summary

AI Engineer with hands-on experience building, evaluating, and improving machine learning systems under real-world constraints. Experience spans **diffusion models**, **RAG pipelines**, **AI agents**, autoregressive generation, and AI automation, applied across financial services, e-commerce, and satellite navigation. Published **5 peer-reviewed papers**. FAU grade **1.2 (Top 1%)**.

Experience

Fraunhofer IIS

Generative AI Research Engineer (Working Student)

Nuremberg, Germany

03/2025 – Present

- Building **DiffGNSS**, a **PyTorch** diffusion model that synthesises **multi-antenna GNSS signals** conditioned on jammer position for **synthetic data augmentation**, including unseen scenarios.
- Iterated on model architecture and loss functions to improve generation quality through reproducible, **MLflow**-tracked experiments and ablations.
- Designed a domain-specific **evaluation framework** to measure model quality where standard metrics fall short, collaborating with signal-processing experts to ground it in physical constraints.
- Extending the model into a **jamming removal** tool that subtracts its predicted jammer from the received signal to recover the clean GNSS signal.

August-Wilhelm Scheer Institut

Applied Machine Learning Engineer (Working Student)

Saarbrücken, Germany

12/2023 – 02/2025

- Built end-to-end ML pipelines predicting **e-commerce returns** on cold-start products with no transaction history, reaching **86% balanced accuracy** under heavy class imbalance and feature sparsity.
- Implemented and tuned **TabNet**, outperforming traditional ML baselines including **XGBoost** and **SVM**, using tabular encoding strategies.
- Conducted feature importance analysis with **SHAP** to identify key drivers of returns, enabling interpretable and actionable recommendations for business stakeholders.
- Refactored and parallelised preprocessing pipelines for a **5x runtime speedup**, and contributed optimised **inference scripts** for the institute's final application.

Hightech Payment Systems

Development Engineer

Casablanca, Morocco

06/2023 – 08/2023

- Enhanced **PowerCARD**, HPS's global payment switching and card management platform, to meet **VISA** and **Mastercard** compliance requirements.
- Worked across **SQL** databases, **CI/CD** pipelines, **Docker**, and **Linux** in a mission-critical, regulated payments environment.

Al Akhawayn University

Applied NLP Research Engineer (Part-time)

Ifrane, Morocco

09/2022 – 05/2023

- Pre-trained two compact transformer language models (**DarELECTRA** 52M, **DarRoBERTa** 80M parameters) from scratch for low-resource Moroccan Darija on a 1 GB code-mixed corpus.
- Fine-tuned the models on two tasks: Darija summarization (**ROUGE-1 19.25 / ROUGE-L 18.01**) beating **state-of-the-art** Arabic models, and topic classification (**86% accuracy / 84% F1**).
- Contributed to a follow-up study on offensive language detection (**90% accuracy / 85% F1**) that stress-tested model robustness with **metamorphic testing** and adversarial attacks.

Wenov: Attijariwafa Bank Innovation Lab

Applied ML Engineer (Intern)

Casablanca, Morocco

05/2022 – 07/2022

- Built a **transformer-based** intent classification system to automatically route **multilingual** client inquiries to the right departments, replacing a rule-based system.
- Outperformed traditional ML baselines by **+7% accuracy / +4% F1**, validated across multiple language variants and edge cases.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg

M.Sc. Artificial Intelligence

Erlangen, Germany

10/2023 – 06/2026

- Grade: 1.2 (Top 1%, German grading scale)
- Thesis: *Finetuning Visual Autoregressive Models for Controllable Image Generation*. First spatial control framework for scale-wise autoregressive text-to-image generation, a **ControlNet**-style model adding six control types over a frozen **2.5B-parameter** backbone. Details and samples: saaddriouech.com/thesis.
- Relevant courses: Generative Models for Signal Processing, Advanced Topics in Deep Learning, Pattern Recognition, Machine Learning for Time Series, Advanced Programming Techniques.

Al Akhawayn University

B.Sc. Computer Science

Ifrane, Morocco

09/2018 – 07/2022

- GPA: 3.83 / 4.0 (Summa Cum Laude)
- Bachelor Thesis: *A Deep Learning-Based Speech Synthesis for Moroccan Darija*. First such system for the dialect, built on **FastSpeech 2** and a **HiFi-GAN** vocoder, reaching **MOS 3.905** on just 2 hours of training data.
- Relevant courses: Artificial Intelligence, Probability & Statistics, Linear Algebra, Data Analysis, Intro. to Big Data.

Personal Projects

ApplAI

2026

- Automated job-application pipeline that scrapes job boards, scores role fit with an LLM, and generates tailored LaTeX CVs and cover letters, delivered to Discord for human approve/reject review.
- Built with **n8n** cron orchestration, **FastAPI**, **Gemini 2.5 Flash** scoring, **Claude** document generation, **Docker Compose**, and a security-first design (JD sanitization, **Ed25519** verification, parameterized SQL).

PaperLens

2026

- RAG system answering natural-language questions across PDFs, Markdown, and web pages with cited sources, using diversity-aware selection so every source is represented in each answer.
- Built with two-stage retrieval (bi-encoder plus **cross-encoder reranking**), a **Qdrant** vector store, **sentence-transformers**, **Groq** (Llama 3.3 70B), **FastAPI**, and an **MRR / Recall@k** evaluation pipeline.

MotiGen

2026

- Agentic pipeline that turns a CV and job description into a tailored motivation letter, parsing both documents and researching the company via live web search before generating.
- Built with **LangGraph** StateGraph, multi-LLM support (**Claude**, **Groq**, **OpenAI**), **Pydantic** structured parsing, **Tavily** search, and a **FastAPI** backend with SSE streaming.

Publications

- **Moroccan Darija Text-to-Speech Synthesis** — *ICDTA 2025*
- **Towards Waste Reduction in E-Commerce: Garment Returns Prediction** — *SN Computer Science, Springer*, 05/2025
- **Strategic Returns Prevention in E-Commerce: Agent-Based Modeling** — *ICAART 2025*
- **Investigating Offensive Language Detection in a Low-Resource Setting** — *Big Data & Cognitive Computing, MDPI*, 11/2024
- **Compact Transformer-based Language Models for Moroccan Darija** — *IEEE CiSt*, 12/2023

Technical Skills

Generative & Deep Learning: PyTorch, TensorFlow, Diffusion Models, Transformers, Hugging Face, Fine-tuning (LoRA/PEFT), Synthetic Data Generation

LLM & Agentic Systems: LangChain, LangGraph, RAG, cross-encoder reranking, Qdrant, LlamaIndex, structured outputs (Pydantic), OpenAI/Anthropic/Groq APIs

Classical ML & Evaluation: Scikit-learn, XGBoost, SHAP, MLflow, TensorBoard

Backend & MLOps: Python, FastAPI, SSE streaming, Streamlit, Docker, n8n, Apache Airflow, CI/CD, Git, SQL, Linux

NLP & Speech: spaCy, TTS (FastSpeech 2, HiFi-GAN), low-resource language modeling

Languages: English (Fluent), French (Advanced), Arabic (Native), German (Learning)